1.45 Matrices: Modelling

Worked Solutions

1 a)

Let x be the number of Wolves on 1st January 2022.

Let y be the number of Foxes on 1st January 2022.

Let z be the number of Bears on 1st January 2022.

1 b)

$$y = x + 100 \Rightarrow x - y = -100.$$

1 c)

$$x + y + z = 1500.$$

1 d)

Number of Wolves increased by 3%: $(1 + 0.03)x = 1.03x$.

Number of Foxes decreased by 5%: $(1 - 0.05)y = 0.95y$.

Number of Bears increased by 2%: $(1 + 0.02)z = 1.02z$.

1 e)

Total population increased by 1%: $(1 + 0.01) \times 1500 = 1.01 \times 1500 = 1515$.

$$\therefore 1.03x + 0.95y + 1.02z = 1515.$$

2 a)

$$M = \begin{pmatrix} 25 & 15 & 10 \\ 30 & 20 & 12 \\ 28 & 18 & 14 \end{pmatrix}, \quad v = \begin{pmatrix} 3.00 \\ 4.50 \\ 2.50 \end{pmatrix}$$

$$Mv = \begin{pmatrix} 25 & 15 & 10 \\ 30 & 20 & 12 \\ 28 & 18 & 14 \end{pmatrix} \begin{pmatrix} 3.00 \\ 4.50 \\ 2.50 \end{pmatrix} = \begin{pmatrix} 167.50 \\ 210.00 \\ 200.00 \end{pmatrix}.$$

2 b)

Total cost = £167.50 + £210.00 + £200.00 = £577.50.

3)

$$M = \begin{pmatrix} 0.25 & 0.30 & 0.28 & 0.20 \\ 0.15 & 0.12 & 0.10 & 0.08 \\ 0.40 & 0.45 & 0.42 & 0.35 \end{pmatrix}, \quad v = \begin{pmatrix} 31 \\ 31 \\ 28 \\ 31 \end{pmatrix}$$

$$Mv = \begin{pmatrix} 0.25 & 0.30 & 0.28 & 0.20 \\ 0.15 & 0.12 & 0.10 & 0.08 \\ 0.40 & 0.45 & 0.42 & 0.35 \end{pmatrix} \begin{pmatrix} 31 \\ 31 \\ 28 \\ 31 \end{pmatrix} = \begin{pmatrix} 31.09 \\ 13.65 \\ 48.96 \end{pmatrix}.$$

So we expect a total of $31.09 \approx 31$ snowy days in Toronto, $13.65 \approx 14$ in Vancouver, and $48.96 \approx 49$ in Montreal.

4 a)

$$M = \begin{pmatrix} 120 & 150 & 140 \\ 135 & 160 & 130 \\ 110 & 140 & 125 \\ 145 & 155 & 135 \end{pmatrix}, \quad v = \begin{pmatrix} 2.50 \\ 2.75 \\ 3.00 \end{pmatrix}.$$

4 b)

$$Mv = \begin{pmatrix} 120 & 150 & 140 \\ 135 & 160 & 130 \\ 110 & 140 & 125 \\ 145 & 155 & 135 \end{pmatrix} \begin{pmatrix} 2.50 \\ 2.75 \\ 3.00 \end{pmatrix} = \begin{pmatrix} 1132.5 \\ 1167.5 \\ 1035 \\ 1193.75 \end{pmatrix}.$$

4 c)

So Driver A accrues a delivery fee of £1132.50, Driver B accrues a delivery fee of £1167.50, Driver C accrues a delivery fee of £1035.00, and Driver D accrues a delivery fee of £1193.75. So the total delivery fee is £4528.75.

5)

Let x be the number of smartphones.

Let y be the number of laptops.

Let z be the number of tablets.

The total number in the warehouse was 600:

$$x + y + z = 600.$$

Originally there were 200 more tablets than smartphones:

$$x - z = -200.$$

Smartphones increased by 3%, laptops increased by 5%, tablets decreased by 8%, total number decreased by 11:

$$1.03x + 1.05y + 0.92z = 589.$$

So:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \\ 1.03 & 1.05 & 0.92 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 600 \\ -200 \\ 589 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \\ 1.03 & 1.05 & 0.92 \end{pmatrix}^{-1} \begin{pmatrix} 600 \\ -200 \\ 589 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 7 & \frac{13}{15} & -\frac{20}{3} \\ -13 & -\frac{11}{15} & \frac{40}{3} \\ 7 & -\frac{2}{15} & -\frac{20}{3} \end{pmatrix} \begin{pmatrix} 600 \\ -200 \\ 589 \end{pmatrix} = \begin{pmatrix} 100 \\ 200 \\ 300 \end{pmatrix}.$$

So originally there were 100 smartphones, 200 laptops, and 300 tablets.

6)

Let x be the amount invested in TechCorp.

Let y be the amount invested in FinCorp.

Let z be the amount invested in RetailCo.

The total amount invested was 5500:

$$x + y + z = 5500.$$

Originally there was £1000 more invested in FinCorp than in TechCorp:

$$x - y = -1000.$$

TechCorp's shares increased by 2.5%, FinCorp's shares decreased by 1.8%, RetailCo's shares increased by 3.2%, and the total value of the investments increased by £12:

$$1.025x + 0.982y + 1.032z = 5512.$$

So:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1.025 & 0.982 & 1.032 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5500 \\ -1000 \\ 5512 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1.025 & 0.982 & 1.032 \end{pmatrix}^{-1} \begin{pmatrix} 5500 \\ -1000 \\ 5512 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{344}{19} & \frac{50}{57} & -\frac{1000}{57} \\ \frac{344}{19} & -\frac{7}{57} & -\frac{1000}{57} \\ -\frac{669}{19} & -\frac{43}{57} & \frac{2000}{57} \end{pmatrix} \begin{pmatrix} 5500 \\ -1000 \\ 5512 \end{pmatrix} = \begin{pmatrix} 2000 \\ 3000 \\ 500 \end{pmatrix}.$$

So originally Sidnie invested £2000 in TechCorp, £3000 in FinCorp, and £500 in RetailCo.

7)

Let x be the number of Adélie penguins.

Let y be the number of Gentoo penguins.

Let z be the number of Chinstrap penguins.

The total number of penguins in the colony was 1350:

$$x + y + z = 1350.$$

Originally there were 210 more Gentoo penguins than Chinstrap penguins:

$$y - z = 210.$$

Adélie penguins decreased by 5%, Gentoo penguins decreased by 10%, Chinstrap penguins increased by 12%, while the total number decreased by 48:

$$0.95x + 0.9y + 1.12z = 1302.$$

So:

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0.95 & 0.9 & 1.12 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1350 \\ 220 \\ 1302 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0.95 & 0.9 & 1.12 \end{pmatrix}^{-1} \begin{pmatrix} 1350 \\ 220 \\ 1302 \end{pmatrix} = \begin{pmatrix} 640 \\ 460 \\ 250 \end{pmatrix}.$$

So originally there were 640 Adélie penguins, 460 Gentoo penguins, and 250 Chinstrap penguins.

8)

Let x be the number of Fiction books sold in 2022.

Let y be the number of Non-Fiction books sold in 2022.

Let z be the number of Children's books sold in 2022.

The total number of books sold in 2023:

$$0.7x + 1.3y + 1.2z = 155800 \quad (1)$$

The total profit in 2022:

$$3x + 4y + 2.5z = 443000 \quad (2)$$

The total profit in 2023:

$$0.95 \times 443000 = 420850.$$

The profit from each genre:

$$3(0.7x) + 4(1.3y) + 2.5(1.2z) = 420850$$

$$\Rightarrow 2.1x + 5.2y + 3z = 420850 \quad (3)$$

So:

$$\begin{pmatrix} 0.7 & 1.3 & 1.2 \\ 3 & 4 & 2.5 \\ 2.1 & 5.2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 155800 \\ 443000 \\ 420850 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0.7 & 1.3 & 1.2 \\ 3 & 4 & 2.5 \\ 2.1 & 5.2 & 3 \end{pmatrix}^{-1} \begin{pmatrix} 155800 \\ 443000 \\ 420850 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{200}{613} & \frac{468}{613} & -\frac{310}{613} \\ -\frac{750}{613} & -\frac{84}{613} & \frac{370}{613} \\ \frac{1440}{613} & -\frac{182}{613} & -\frac{220}{613} \end{pmatrix} \begin{pmatrix} 155800 \\ 443000 \\ 420850 \end{pmatrix} = \begin{pmatrix} 74552.20228 \\ 2695.758564 \\ 83424.14356 \end{pmatrix}.$$

So in 2022, 74600 Fiction books, 2700 Non-Fiction books, and 83400 Children's books were sold (3 SF).

9 a)

P_0 is the initial size of the population of puffins on the island.

9 b)

P_{10} is the size of the population of puffins on the island on 1st January 2025.

10)

$$A_1 = 0.51A_0 + 0.32B_0 = 0.51(2.5) + 0.32(0.9) = 1.563.$$

11)

$$C_2 = 0.63C_1 + 0.28D_0 = 0.63(2.7) + 0.28(1.5) = 2.121.$$

12)

The number of pups that survive from year n to year $n+1$ is given by $0.83P_n$.

The number of pups that transition to adulthood each year is $\frac{1}{4}(0.83P_n)$.

So the remaining pup population is year $n+1$ is the surviving pups minus those that transitioned to adulthood:

$$P_{n+1} = 0.83P_n - \frac{1}{4}(0.83P_n) = 0.6225P_n.$$

13)

As adult fish survive at a rate of 0.88 and one fifth of the surviving larvae mature into adults each year, we must have that $F_{n+1} = 0.88F_n + 0.2L_n$.

As the larvae population survives at a rate of 0.75 each year, we must have that $L_{n+1} = 0.75L_n$.

So:

$$\begin{pmatrix} F_{n+1} \\ L_{n+1} \end{pmatrix} = \begin{pmatrix} 0.88 & 0.2 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} F_n \\ L_n \end{pmatrix}.$$

14 a)

The top row of the matrix says that 95% of the worker bees survive each year, and 15% of the surviving larvae mature into workers.

14 b)

The bottom row of the matrix says that 80% of the larvae survive each year.

14 c)

$W_0 = 4000$ and $L_0 = 2000$.

$$\begin{pmatrix} W_1 \\ L_1 \end{pmatrix} = \begin{pmatrix} 0.95 & 0.15 \\ 0 & 0.80 \end{pmatrix} \begin{pmatrix} W_0 \\ L_0 \end{pmatrix} = \begin{pmatrix} 0.95 & 0.15 \\ 0 & 0.80 \end{pmatrix} \begin{pmatrix} 4000 \\ 2000 \end{pmatrix} = \begin{pmatrix} 4100 \\ 1600 \end{pmatrix}.$$

On 1st January 2025, the number of worker bees will be 4100 and the number of larvae will be 1600.

14 d)

$$\begin{pmatrix} W_2 \\ L_2 \end{pmatrix} = \begin{pmatrix} 0.95 & 0.15 \\ 0 & 0.80 \end{pmatrix} \begin{pmatrix} W_1 \\ L_1 \end{pmatrix} = \begin{pmatrix} 0.95 & 0.15 \\ 0 & 0.80 \end{pmatrix} \begin{pmatrix} 4100 \\ 1600 \end{pmatrix} = \begin{pmatrix} 4135 \\ 1280 \end{pmatrix}.$$

On 1st January 2026, the number of worker bees will be 4135 and the number of larvae will be 1280.

15 a)

The constant k represents the survival rate of yearlings. It determines the proportion of yearlings that survive to the following year without maturing into adults.

15 b)

Total population at the start of the study:

$$Y_0 + M_0 = 72000.$$

Number of Mature Deer at the start of the study is $M_0 = 55200$.

$$\therefore Y_0 + 55200 = 72000 \Rightarrow Y_0 = 16800.$$

After one year, we have $Y_1 = 15816$ and $M_1 = 46824$.

From the matrix equation:

$$Y_{n+1} = kY_n + 0.18M_n.$$

So $Y_1 = kY_0 + 0.18M_0 \Rightarrow 15816 = k(16800) + 0.18(55200)$.

$$\Rightarrow k = 0.35.$$

15 c)

Using the matrix equation:

$$Y_2 = kY_1 + 0.18M_1 = 0.35 \times 15816 + 0.18 \times 46824 = 13963.92.$$

So the predicted number of yearlings after two years is approximately 14,000 (3 SF).

15 d)

The model could be adjusted by introducing a limiting factor for the yearling birth rate. For example, the birth rate could be modelled as decreasing when the total population exceeds a certain threshold. This could be done by replacing the constant 0.18 with a function that decreases as the population increases, such as

$$0.18 \times \left(1 - \frac{\text{Total population}}{\text{Threshold}}\right).$$

i.e.

$$\begin{pmatrix} Y_{n+1} \\ M_{n+1} \end{pmatrix} = \begin{pmatrix} k & 0.18 \times \left(1 - \frac{\text{Total population}}{\text{Threshold}}\right) \\ 0.06 & 0.83 \end{pmatrix} \begin{pmatrix} Y_n \\ M_n \end{pmatrix}.$$

16 a)

$$M_{n+1} = 0.98M_n + (0.15 \times 0.74)S_n = 0.98M_n + 0.111S_n.$$

16 b)

$$S_{n+1} = 0.16M_n + (1 - 0.15) \times 0.74S_n = 0.16M_n + 0.629S_n.$$

16 c)

$$\begin{pmatrix} M_0 \\ S_0 \end{pmatrix} = \begin{pmatrix} 1.4 \\ 0.6 \end{pmatrix}.$$

16 d)

$$T = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix}.$$

16 e)

i)

$$\begin{pmatrix} M_0 \\ S_0 \end{pmatrix} = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix} \begin{pmatrix} M_{-1} \\ S_{-1} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} M_{-1} \\ S_{-1} \end{pmatrix} = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix}^{-1} \begin{pmatrix} M_0 \\ S_0 \end{pmatrix}$$

$$= \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix}^{-1} \begin{pmatrix} 1.4 \\ 0.6 \end{pmatrix} = \begin{pmatrix} \frac{850}{809} & -\frac{150}{809} \\ -\frac{8000}{29933} & \frac{49000}{29933} \end{pmatrix} \begin{pmatrix} 1.4 \\ 0.6 \end{pmatrix} = \begin{pmatrix} \frac{1100}{809} \\ \frac{18200}{29933} \end{pmatrix} = \begin{pmatrix} 1.35970\ldots \\ 0.60802\ldots \end{pmatrix}.$$

So the number of mature oak trees was 1360 (3 SF) and the number of saplings was 608 (3 SF). In total, there is predicted to be 1970 trees (3 SF). ii)

$$\begin{pmatrix} M_4 \\ S_4 \end{pmatrix} = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix}^4 \begin{pmatrix} M_0 \\ S_0 \end{pmatrix}$$

$$= \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix}^4 \begin{pmatrix} 1.4 \\ 0.6 \end{pmatrix} = \begin{pmatrix} 1.002775512 & 0.248531403 \\ 0.3582434638 & 0.2168789134 \end{pmatrix} \begin{pmatrix} 1.4 \\ 0.6 \end{pmatrix} = \begin{pmatrix} 1.55300\ldots \\ 0.63166\ldots \end{pmatrix}.$$

So the number of mature oak trees is predicted to be 1550 (3 SF) and the number of saplings is predicted to be 632 (3 SF). In total, there is predicted to be 2180 trees (3 SF).

16 f)

$$\begin{pmatrix} M_{n+1} \\ S_{n+1} \end{pmatrix} = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix} \begin{pmatrix} M_n \\ S_n \end{pmatrix} - \begin{pmatrix} 0.1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} M_1 \\ S_1 \end{pmatrix} = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix} \begin{pmatrix} M_0 \\ S_0 \end{pmatrix} - \begin{pmatrix} 0.1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix} \begin{pmatrix} 1.4 \\ 0.6 \end{pmatrix} - \begin{pmatrix} 0.1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1.3386 \\ 0.6014 \end{pmatrix}$$

$$\begin{pmatrix} M_2 \\ S_2 \end{pmatrix} = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix} \begin{pmatrix} M_1 \\ S_1 \end{pmatrix} - \begin{pmatrix} 0.1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix} \begin{pmatrix} 1.3386 \\ 0.6014 \end{pmatrix} - \begin{pmatrix} 0.1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1.2785834 \\ 0.5924566 \end{pmatrix}$$

$$\begin{pmatrix} M_3 \\ S_3 \end{pmatrix} = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix} \begin{pmatrix} M_2 \\ S_2 \end{pmatrix} - \begin{pmatrix} 0.1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix} \begin{pmatrix} 1.2785834 \\ 0.5924566 \end{pmatrix} - \begin{pmatrix} 0.1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1.218774415 \\ 0.5772285454 \end{pmatrix}$$

$$\begin{pmatrix} M_4 \\ S_4 \end{pmatrix} = \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix} \begin{pmatrix} M_3 \\ S_3 \end{pmatrix} - \begin{pmatrix} 0.1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0.98 & 0.111 \\ 0.16 & 0.629 \end{pmatrix} \begin{pmatrix} 1.218774415 \\ 0.5772285454 \end{pmatrix} - \begin{pmatrix} 0.1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1.158471295 \\ 0.5580806614 \end{pmatrix}.$$

So the number of mature oak trees is predicted to be 1160 (3 SF) and the number of saplings is 558 (3 SF). In total, there is predicted to be 1720 trees (3 SF).